

# Haocheng He

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## Professional Summary

Aviation Engineering MSc candidate at PolyU (GPA: 3.95/4.3), with an IE undergraduate and Law minor from GDUT. Experience spans airport simulation (AnyLogic, 5 routing algorithms), manufacturing optimization (line balance 58% → 70%), and Ascend 310B hardware R&D. 3 patent applications.

## Education

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| <b>The Hong Kong Polytechnic University</b><br><i>Master of Science in Aviation Engineering and Operations Management</i><br>GPA: 3.95 / 4.3 (Current)<br>Thesis: Simulation Modelling and Optimization for Airport Ferry Vehicle Scheduling under Uncertainty                                                                   | Hong Kong SAR, China<br>Sep 2025 – Jan 2027 |
| <b>Guangdong University of Technology</b><br><i>Bachelor of Engineering in Industrial Engineering</i><br>GPA: 3.67 (Top 20%)<br>Thesis: AnyLogic-Based Simulation Optimization of Emergency Vehicle Route Planning in Large Airports<br>Honors: University-level Excellent Graduation Thesis, Graduation Design Innovation Award | Guangzhou, China<br>Sep 2021 – Jun 2025     |
| <i>Bachelor of Laws (Minor Degree in Law)</i><br>GPA: 3.61<br>Thesis: Coordination Between Copyright Protection and Privacy Protection in Digital Rights Management                                                                                                                                                              | Sep 2022 – Jun 2025                         |

## Experience

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| <b>Assistant AI Engineer</b><br><i>Shenzhen Wuxian Hongyin Tech.</i> <ul style="list-style-type: none"><li>Participated in Ascend 310B smart piano R&amp;D across 3 development phases; tested 10+ boards (CPLD, STM32), iterated motor driver 3 times, developed firmware and authored 5 technical documents.</li><li>Participated in camera- and Ascend-based piano fingering recognition module integration (Python/C++); diagnosed cross-layer issues.</li><li>Maintained Ubuntu servers, configured Huawei network equipment, operated 3D printer, supported technical interviews.</li></ul>                  | May 2025 – Jun 2026<br>Shenzhen, China  |
| <b>Research Assistant</b><br><i>GDUT (with Guangzhou Baiyun Int'l Airport)</i> <ul style="list-style-type: none"><li>Built an airport road-network, apron, and runway-area simulation model in AnyLogic, Java, and MySQL for aircraft landing and emergency fire-truck dispatch scenarios.</li><li>Designed single/multi-vehicle, single/multi-event, and runway-road-closure scenarios; compared D* Lite, A*, Dijkstra, BFS, and GA routing algorithms.</li><li>Completed 4 emergency-scenario comparisons across 5 routing algorithms; D* Lite achieved the highest average reliability index (1.342).</li></ul> | Sep 2023 – Jun 2025<br>Guangzhou, China |
| <b>Industrial Engineering Intern</b><br><i>Midea Group — Kitchen &amp; Water Heater Division</i> <ul style="list-style-type: none"><li>Conducted stopwatch time studies across all workstations on a dishwasher sheet-metal line, built operator workload tables, and identified 5 over-cycle steps.</li><li>Proposed 22 improvements using lean methods; simulation showed line balance improving from 58% to 70%, reducing headcount from 20 to 18.</li></ul>                                                                                                                                                    | Apr 2024 – May 2024<br>Foshan, China    |
| <b>Manufacturing Operations Intern</b><br><i>Guangxi LiuGong Machinery Co., Ltd.</i> <ul style="list-style-type: none"><li>Timed 12 workstations on the wheel loader assembly line with a stopwatch; identified 3 major bottlenecks using ECRS analysis.</li><li>Proposed layout adjustments estimated to reduce cycle time by approximately 8% in simulation.</li></ul>                                                                                                                                                                                                                                           | Jul 2023 – Aug 2023<br>Liuzhou, China   |

## Projects

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| <b>Aircraft Engine Remaining Useful Life Prediction (Course Project)</b><br><i>Guangdong University of Technology</i> <ul style="list-style-type: none"><li>Built an RUL prediction workflow on NASA C-MAPSS FD001 multivariate time-series data; cleaned operating-condition data and 21 sensor signals, constructed sequence features, and applied PCA.</li><li>Compared LSTM, BiLSTM, Transformer, and Random Forest baselines for time-series forecasting; after iterative tuning, the BiLSTM model reached about 8% final error from an initial model error of about 50%.</li></ul> | Apr 2024 – Jul 2024<br>Guangzhou |
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## Skills & Intellectual Property

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| <b>Programming:</b> Python, Java, SQL/MySQL, C (Embedded/STM32), JS/TS                                                                                                                                                                                                                                                                                                                                                               |
| <b>ML &amp; Algorithms:</b> LSTM, BiLSTM, Transformer, Random Forest, PCA, Dijkstra, A*, BFS, GA, OR-Tools                                                                                                                                                                                                                                                                                                                           |
| <b>Engineering:</b> AnyLogic, Plant Simulation, Ubuntu/Linux, Git, EDA, Oscilloscope, Logic Analyzer                                                                                                                                                                                                                                                                                                                                 |
| <b>AI:</b> LLM Workflows, Prompt Engineering                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Languages:</b> English (IELTS 6.5), Mandarin (native), Cantonese (conversational)                                                                                                                                                                                                                                                                                                                                                 |
| <b>Patent Applications:</b> <i>MCMC Motion Control for PI Coating Machine Controller</i> (CN122085794A, 2026); <i>Coating Control via Deep Reinforcement Learning</i> (CN121325552A, 2025); <i>Energy Regulation for Household Solar Storage</i> (CN117767437A, 2024). UM: <i>Wire Harness Production Equipment</i> (CN221352455U, 2024). SW: <i>Financial Data Analysis</i> (2024SR0146827); <i>Flood Dispatch</i> (2024SR0549247). |